

Module 4: Supervised Learning
Submodule: 4.7 Naive Baye
Lecture: Prediction and Laplace Smoothing

XCS229 Machine Learning

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I. Introduction

In this lecture, we explore key concepts in predictive modeling, focusing on how to compute the probability $P(y = 1|x)$ for classification. We will introduce **Laplace smoothing** as a solution to the issue of zero probabilities for unseen features. By the end, you will understand how to manage predictions effectively, even with limited data.

II. Probability Computation

We want to compute $P(y = 1|x)$ as per the standard methodology:

$$P(y = 1|x) = \frac{P(x|y = 1)P(y = 1)}{P(x)}$$

If $P(y = 1|x) > 0.5$, we classify the example as positive; if $P(y = 1|x) < 0.5$, we classify as negative.

III. The Zero-Probability Problem

In this section, we discuss a common issue encountered in Naive Bayes classifiers when certain features do not appear in the training dataset, illustrated using the example of the word “aardvark.”

Suppose the word “aardvark” never appears in the training set. As a result, for every example i :

$$x_2^{(i)} = 0, \quad \forall i$$

This indicates that the second feature (word) does not show up in any training example. We estimate probabilities for the feature given different classes:

$$\phi_{2|y=1} = \frac{0}{\# \text{ of positive examples}} = 0 \quad (1)$$

$$\phi_{2|y=0} = \frac{0}{\# \text{ of negative examples}} = 0 \quad (2)$$

To compute the overall probability $P(x)$, we apply the following equation:

$$P(x) = P(x|y = 1)P(y = 1) + P(x|y = 0)P(y = 0)$$

This can be further broken down into:

$$P(x) = \prod_{j=1}^d p(x_j|y = 1)P(y = 1) + \prod_{j=1}^d p(x_j|y = 0)P(y = 0)$$

Given the previous estimations:

$$\begin{aligned} p(x_2 = 1|y = 1) &= \phi_{2|y=1} = 0 \\ p(x_2 = 1|y = 0) &= \phi_{2|y=0} = 0 \end{aligned}$$

Thus, we have:

$$P(x) = 0 \cdot P(y = 1) + 0 \cdot P(y = 0) = 0$$

We conclude that:

$$P(y = 1|x) = \frac{P(x|y = 1)P(y = 1)}{P(x)} = \frac{0}{0}$$

This leads to an indeterminate form $\frac{0}{0}$ when calculating $P(y = 1|x)$.

This situation is problematic because it indicates that our model cannot handle unseen features, leading to zero probabilities. It is possible to observe entirely new words in your test set.

To mitigate this, we use Laplace smoothing, which allows us to introduce a small probability for unseen features. This adjustment indicates that we should not completely rely on the training data. Laplace smoothing proposes that instead of counting zero occurrences, we adjust the counts by adding one (or a small constant) to all counts to ensure no probability is exactly zero.

IV. Laplace Smoothing

To avoid the issue of zero probabilities in our model, we apply **Laplace smoothing**, which allows for a small probability on unseen features. This is crucial for maintaining sensible class predictions.

Simple example:

To estimate the bias of a coin, we model the outcomes of flipping the coin using a Bernoulli distribution. Let Z represent a random variable that indicates the result of a coin flip, where:

- $Z = 1$ corresponds to heads,
- $Z = 0$ corresponds to tails.

This can be mathematically expressed as:

$$Z \sim \text{Bernoulli}(\phi)$$

where ϕ is the unknown probability of getting heads.

Suppose we perform n trials of flipping the coin. The outcomes of these trials can be represented as follows:

$$\begin{array}{llll} Z^{(1)} & \text{tail} & = 0 & (1 - \phi) \\ Z^{(2)} & \text{tail} & = 0 & (1 - \phi) \\ & \vdots & & \\ Z^{(n)} & \text{heads} & = 1 & \phi \end{array}$$

Here, each $Z^{(i)}$ results in a head or tail depending on the respective probability ϕ or $1 - \phi$.

The likelihood function $L(\phi)$, which represents the probability of observing the data given the parameter ϕ , can be formulated as:

$$L(\phi) = (1 - \phi)(1 - \phi) \cdots \phi = (1 - \phi)^t \cdot \phi^h$$

where t is the number of tails and h is the number of heads.

To find the value of ϕ that maximizes the likelihood function, we can take the argument that maximizes $\log L(\phi)$:

$$\arg \max_{\phi} \log L(\phi) = \frac{h}{h + t}$$

This formula indicates that the maximum likelihood estimate of ϕ is simply the ratio of heads observed in the trials. The expression $\frac{h}{h+t}$ provides an empirical estimate of the bias of the coin based on observed data, which is intuitive as it reflects the observed proportion of heads in the experiment.

Another example

Let us consider an example where we flip a coin three times and observe the results:

$$\begin{array}{lll} Z^{(1)} & \text{tail} & = 0 \quad (1 - \phi) \\ Z^{(2)} & \text{tail} & = 0 \quad (1 - \phi) \\ Z^{(3)} & \text{tail} & = 0 \quad (1 - \phi) \end{array}$$

The standard estimate for ϕ would be:

$$\phi = \frac{h}{h + t} = \frac{0}{0 + 3} = 0.$$

However, is it reasonable to conclude that the coin never produces heads? This is where prior knowledge can play a crucial role. To address this limitation, we can use Laplace smoothing, which incorporates a prior into our estimate. The formula for **Laplace smoothing** is:

$$\phi = \frac{h + 1}{h + t + 2}$$

Thus,

$$\phi = \frac{h + 1}{h + t + 2} = \frac{0 + 1}{0 + 3 + 2} = \frac{1}{5}$$

This ensures $\phi > 0$ even when no heads are observed.

Large Data Example:

Now, let's consider a larger dataset where:

$$h = 100, \quad t = 60$$

Using the standard method, the estimate for ϕ is:

$$\phi = \frac{100}{100 + 60} = \frac{100}{160} = 0.625$$

Applying Laplace smoothing gives us:

$$\phi = \frac{100 + 1}{100 + 60 + 2} = \frac{101}{162} \approx 0.623$$

Both estimates are quite similar because the data sizes dominate the influence of the +1 in the numerator and the +2 in the denominator.

Note:

- In cases with limited trials (such as three), one might question the reliability of a ϕ estimate of 0. It's possible, through chance, to observe only tails. A fair coin will typically produce heads as well, so one should consider this when analyzing results.
- Laplace smoothing allows our estimation to be less extreme. By giving our estimate a slight adjustment (adding 1 to heads and 2 to the total count), we acknowledge that while our data may show tails, we still account for the possibility of heads existing, leading to a more reasonable estimate.
- Laplace smoothing is particularly useful when data is sparse. As the amount of data increases, the influence of Laplace smoothing diminishes, and the estimates converge toward the true proportions observed in the data. When large datasets are available, the raw counts become more trustworthy.

Laplace smoothing parameters

From our Naive Bayes classifier, we know that

$$\begin{aligned}\phi_{j|y=1} &= \frac{\sum_{i=1}^n \mathbb{1}\{x_j^{(i)} = 1, y^{(i)} = 1\}}{\sum_{i=1}^n \mathbb{1}\{y^{(i)} = 1\}} \\ \phi_{j|y=0} &= \frac{\sum_{i=1}^n \mathbb{1}\{x_j^{(i)} = 1, y^{(i)} = 0\}}{\sum_{i=1}^n \mathbb{1}\{y^{(i)} = 0\}}\end{aligned}$$

With Laplace smoothing, the estimations of the parameters adjust as follows:

$$\begin{aligned}\phi_{j|y=1} &= \frac{1 + \sum_{i=1}^n \mathbb{1}\{x_j^{(i)} = 1, y^{(i)} = 1\}}{2 + \sum_{i=1}^n \mathbb{1}\{y^{(i)} = 1\}} \\ \phi_{j|y=0} &= \frac{1 + \sum_{i=1}^n \mathbb{1}\{x_j^{(i)} = 1, y^{(i)} = 0\}}{2 + \sum_{i=1}^n \mathbb{1}\{y^{(i)} = 0\}}\end{aligned}$$

From the case where the feature does not appear in the dataset in equations (1, 2). With Laplace smoothing, however, the estimation changes:

$$\begin{aligned}\phi_{2|y=1} &= \frac{0 + 1}{\# \text{ of positive examples} + 2} > 0 \\ \phi_{2|y=0} &= \frac{0 + 1}{\# \text{ of negative examples} + 2} > 0\end{aligned}$$

Extension to Multiple Classes

If extending this idea to multiple classes (such as rolling a die):

Given $z \in \{0, 1, \dots, k-1\}$, the probability without smoothing is:

$$p(z = j) = \frac{\sum_{i=1}^n \mathbb{1}\{z^{(i)} = j\}}{n}$$

After applying Laplace smoothing, it would be:

$$p(z = j) = \frac{1 + \sum_{i=1}^n \mathbb{1}\{z^{(i)} = j\}}{n + k}$$

This adjustment adds uniformity and reliability to estimates across various categories.

Application of Laplace Smoothing

When applied to our classification problem (such as spam filtering), the numerator counts the occurrences of specific features in positive examples, while the denominator represents the total number of positive examples. This ensures no probability estimate is exactly zero, addressing the issue of extreme estimates (like for the ‘aardvark’) effectively.

Laplace smoothing effectively makes sure that if a class has not appeared in the dataset, it won’t lead to a zero estimate:

- It adjusts all estimates to be greater than zero.
- Helps ensure that models remain robust, particularly in cases with few examples.

Laplace smoothing provides a means to moderate the estimates derived from limited data by introducing a small constant. It ensures that all feature probabilities are non-zero, enhancing the model's ability to generalize from sparse datasets.

V. Summary

In this lecture, we have explored essential concepts related to predictive modeling and the challenges faced in calculating probabilities using Naive Bayes classifiers. Here are the key takeaways:

- We focused on the computation of conditional probabilities, particularly $P(y = 1|x)$, and how classifications depend on these probabilities.
- The zero-probability problem arises when features are unseen in the training data, leading to indeterminate forms such as $\frac{0}{0}$. This limitation impairs effective classification.
- We introduced **Laplace smoothing** as a solution to the zero-probability problem. By adding a small constant (usually 1) to counts, we ensured that no probability estimate is exactly zero.
- Laplace smoothing not only mitigates the issues arising from sparse data but also provides more reliable probability estimates that can handle unseen features.
- We demonstrated Laplace smoothing with various examples, including flipping a coin and applying it within the context of Naive Bayes classifiers.
- Finally, we extended the concept of Laplace smoothing to scenarios involving multiple classes, thereby enhancing the robustness and versatility of our probabilistic models.